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ESP32 RFID Anti-Wall-Jumping System

Advanced multi-sensor security solution for perimeter protection

🔗 Overview

The **ESP32 RFID Anti-Wall-Jumping System** is a robust, real-time perimeter security prototype designed to detect unauthorized access (like wall climbing or jumping) and respond instantly. It leverages an Active IR Beam sensor for reliable perimeter detection, a long-range UHF RFID reader for authorized access, a DFPlayer Mini for audio alarms, and a SIM800L module for instant SMS notifications to security personnel.

The system features a **built-in Captive Portal (Web UI)** for easy configuration on the fly, avoiding the need to re-flash the firmware to change settings.

🔑 Key Features

- 🔦 **Active Perimeter Detection:** Uses an Active Single-Beam IR Detector for extreme weather-resistant and long-range intrusion detection (replacing old ultrasonic sensors).
- 📠 **Instant SMS Alerts:** Uses SIM800L to send immediate notifications to a registered emergency contact upon intrusion.
- 🔊 **Audio Alarms:** Integrates DFPlayer Mini to broadcast custom localized voice alarms or sirens.
- 📶 **Long Range RFID (UHF WG26):** Allows authorized personnel (using the RD906M Directional Panel Antenna) to bypass the system from a distance.
- 💾 **Local SD Card Logging:** All events and access attempts are saved to an onboard MicroSD card in FAT32 format.
- 🌐 **Web-Based Captive Portal:** Built-in async web server allowing administrators to configure system thresholds, modes, and sensors over Wi-Fi without compiling code.
- 🕒 **Real-Time Clock:** Built-in DS3231 RTC for accurate time-stamping of intrusion logs.
- 🔒 **Hardened Firmware:** Production-grade concurrency protection with dedicated FreeRTOS mutexes, deferred restart handling, validated RTC input, and guarded SD operations.

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🏗️ System Architecture

System Architecture



Hardware Stack

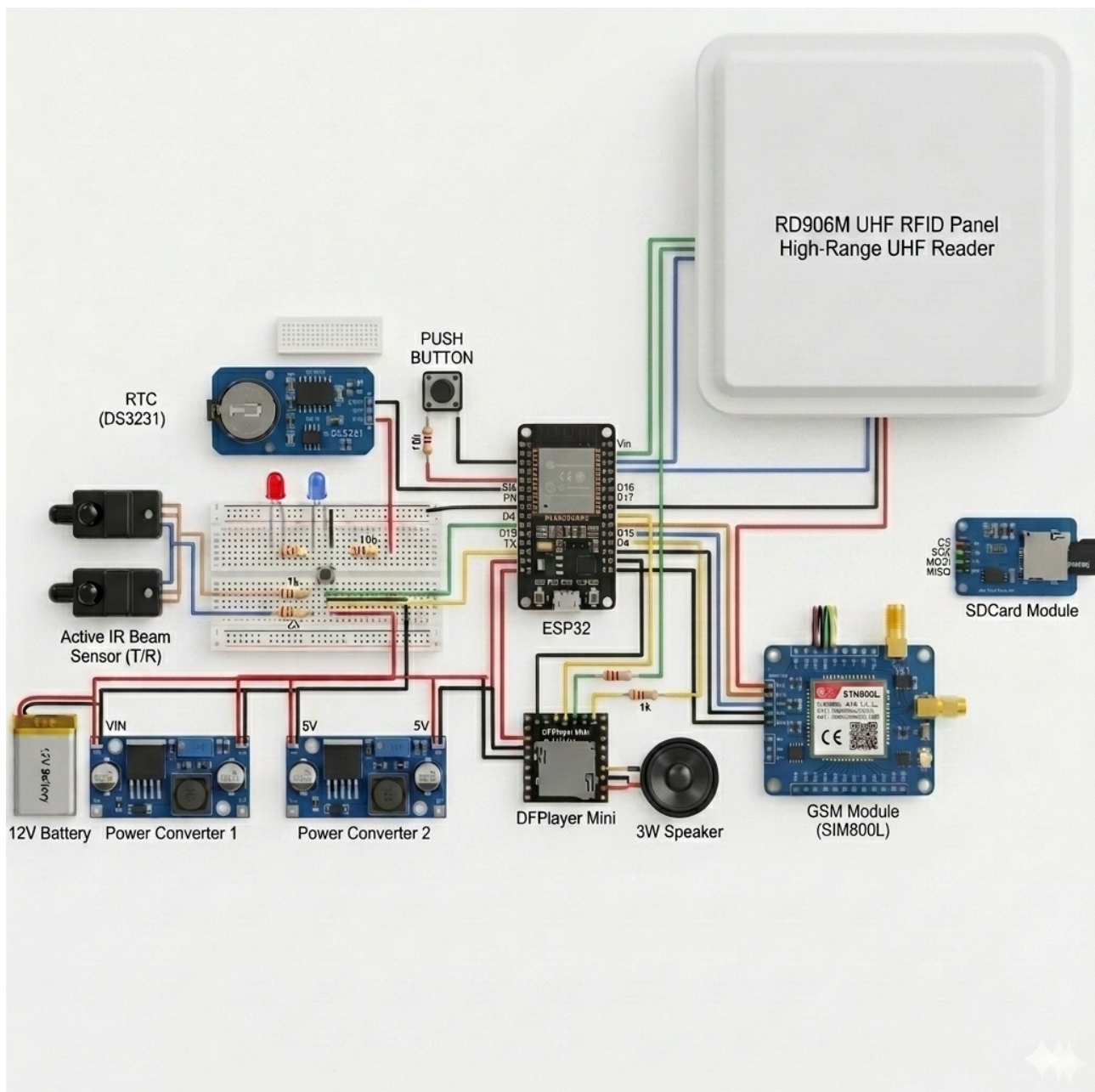
Component	Function
ESP32-38P	Core processing and Web Server AP
RD906M UHF Panel	Long-range RFID Scanning
Active IR Beam	Perimeter breach / motion detection
SIM800L	GSM Module for SMS alerts
DFPlayer Mini	MP3 Audio playback
DS3231	Precision RTC for logging
MicroSD Module	Local event storage

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Realistic Component Layout

[NOTE] Visual Reference Only

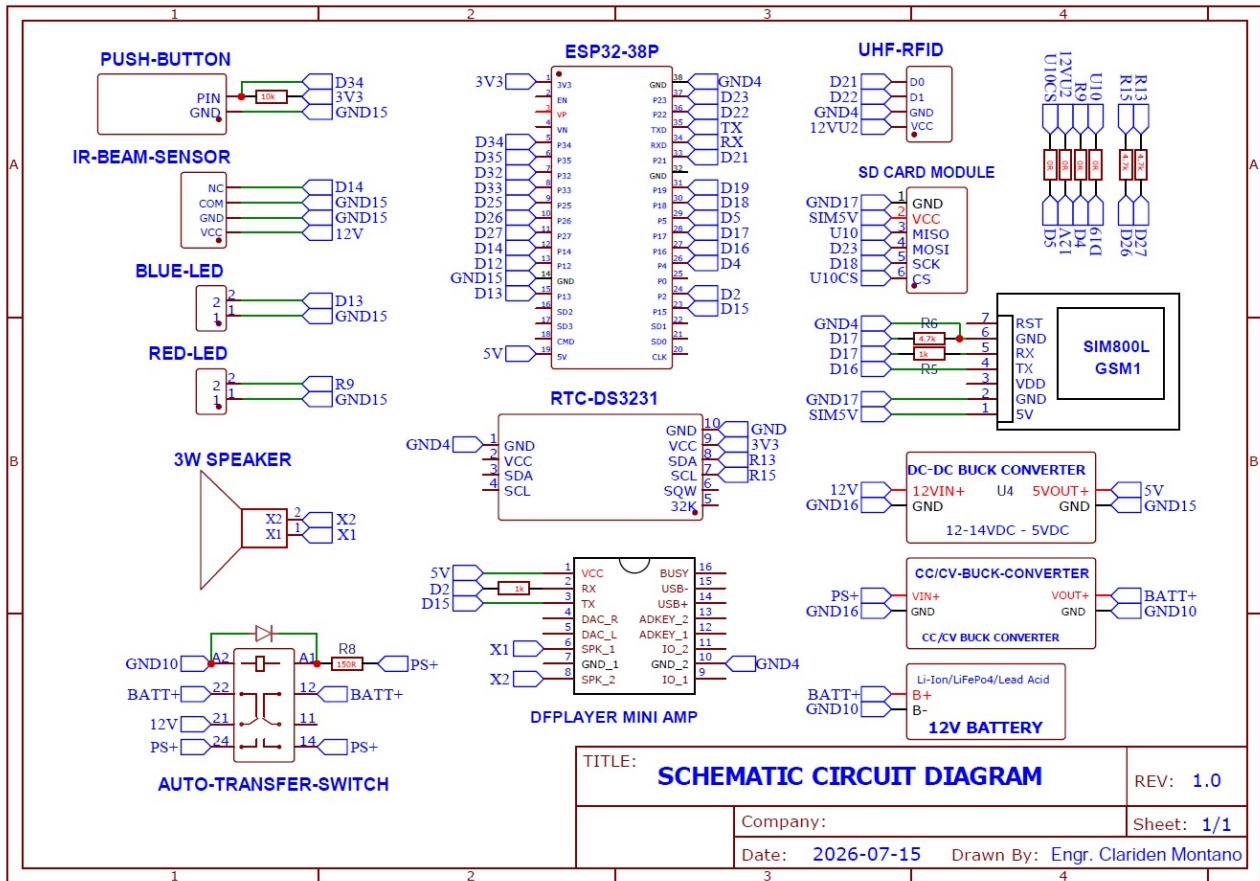
The realistic component layout below is for visual purposes and aesthetic representation. For exact pin mapping and electrical connections, strictly follow the official Schematic Circuit Diagram.



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🔧 Schematic Circuit Diagram

For full hardware wiring instructions and component mapping, please refer to the official wiring schematic:



🔗 Optimized Pin Assignment (ESP32-38P)

All peripheral pairs are assigned to physically adjacent GPIO pins on the ESP32-38P to minimize wire crossings on a prototype board.

LEFT SIDE (top → bottom)	RIGHT SIDE
GPIO34 ← MODE BUTTON	GPIO23 ← SPI MOSI
GPIO35 ← (Unassigned)	GPIO22 ← WIEGAND D1 } RD906M
GPIO32 ← (Unassigned)	GPIO21 ← WIEGAND D0 } Reader
GPIO33 ← (Unassigned)	GPIO19 ← SPI MISO
GPIO25 ← (Unassigned)	GPIO18 ← SPI SCK } SD Card
GPIO26 ← RTC SCL	GPIO5 ← SD CS
GPIO27 ← RTC SDA } I2C	GPIO17 ← SIM800 TX
GPIO14 ← IR BEAM	GPIO16 ← SIM800 RX } UART1
GPIO12 ← (Unassigned)	GPIO4 ← Alarm LED (Red)
GND	GPIO0 ← (Unused, boot clock)
GPIO13 ← Status LED (Blue)	GPIO2 ← DFPlayer TX } UART2
	GPIO15 ← DFPlayer RX

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🎮 2. How the System Works

The system has two main states or "modes". You can switch between them by holding down the **Mode Button** (GPIO 34).

2.1 Normal Mode (Guarding)

- **Blue Light:** Stays ON.
- **What it does:** The system uses the IR beam to watch the wall. If the beam is broken (meaning someone is jumping the wall), it triggers the alarm.
- **When triggered:** It plays a warning sound, flashes the red light, and sends a text message to a security guard.

2.2 Setup Mode

- **Blue Light:** *Blinking fast.*
- **How to enter:** *Press and hold the Mode Button for 3 seconds.*
- **What it does:** *The system stops watching the wall and creates its own Wi-Fi network. You can connect to this Wi-Fi to change settings using your phone or laptop.*
- **How to exit:** *Press and hold the button for 3 seconds again. (It will also exit automatically after 1 hour if you forget).*

2.3 Remote Control via Text Message (SMS)

You don't always have to press the physical button to change modes! You can send a text message to the system's SIM card to control it remotely:

- Text AP ON: *Turns on the Wi-Fi Setup Mode so you can connect to the dashboard.*
- Text AP OFF: *Turns off the Wi-Fi Setup Mode and goes back to Normal Guarding mode.*

(Note: For extra security, if "Number Lock" is enabled, the system will only obey text commands sent from the registered security guard's phone number!)

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3. Changing Settings on the Dashboard

3.1 Connecting to the Setup Page

1. Put the system into **Setup Mode** (hold the button until the blue light blinks).
2. On your phone or laptop, connect to the Wi-Fi network:
 - **Wi-Fi Name:** AntiWallJump-Config
 - **Password:** configure123
3. A login page should pop up. If it doesn't, open a web browser and go to <http://192.168.4.1>.
4. The default admin password is admin.

3.2 What's on the Dashboard?

The setup page has different sections and buttons to help you control the system:

Top Menu Buttons (Quick Actions):

- **Sync Clock (Clock Icon):** *Click this to instantly set the system's time to match your phone or computer.*
- **Dark/Light Mode (Moon/Sun Icon):** *Switch the screen between dark and light colors.*
- **Logout (Door Icon):** *Securely sign out of the setup page.*
- **Stop Portal (Power Icon):** *Quickly exit setup mode and turn the alarm back on right away.*

Bottom Navigation Tabs:

- **Dashboard:** *See the "Node Overview" (system health like the memory card status) and check the overall security status.*
- **Registry:** *Register student ID cards and see who triggered the alarm.*
- **Settings:** *Change the phone number for alerts, choose how close someone has to be to trigger the alarm, and pick which sound plays.*
- **Records:** *View a history of all past alarms and download a record file.*

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4. Student ID Cards (Registry)

The system keeps track of authorized students using their UHF RFID ID cards.

4.1 Adding a New Student Card

1. Go to the **Registry** tab on the dashboard.
2. Click **Scan Card**.
3. Tap a new ID card on the UHF RFID panel. The card's special ID will show up on the screen.
4. Type in the student's name and school ID number.
5. Click **Register** to save them in the system.

4.2 What happens when the alarm triggers?

- If the sensors spot someone climbing, it logs an **UNKNOWN INTRUDER**.
 - If someone holds a registered student ID card to the reader, the system grants them safe passage.
 - If an unregistered card is tapped, it triggers the alarm to keep the area secure.
-

5. Alarm Records

The system saves a history of every time the alarm goes off. It saves this inside its internal memory and on a MicroSD card, so nothing gets lost!

5.1 Downloading the Records

You can download a spreadsheet file of all the past alarms:

1. Go to the **Records** tab on the dashboard.
2. Click "**Download Logs**".
3. A file called `Campus_Security_Logs.csv` will download to your device. You can open this with Excel or Google Sheets to see a neat list of everything that happened.